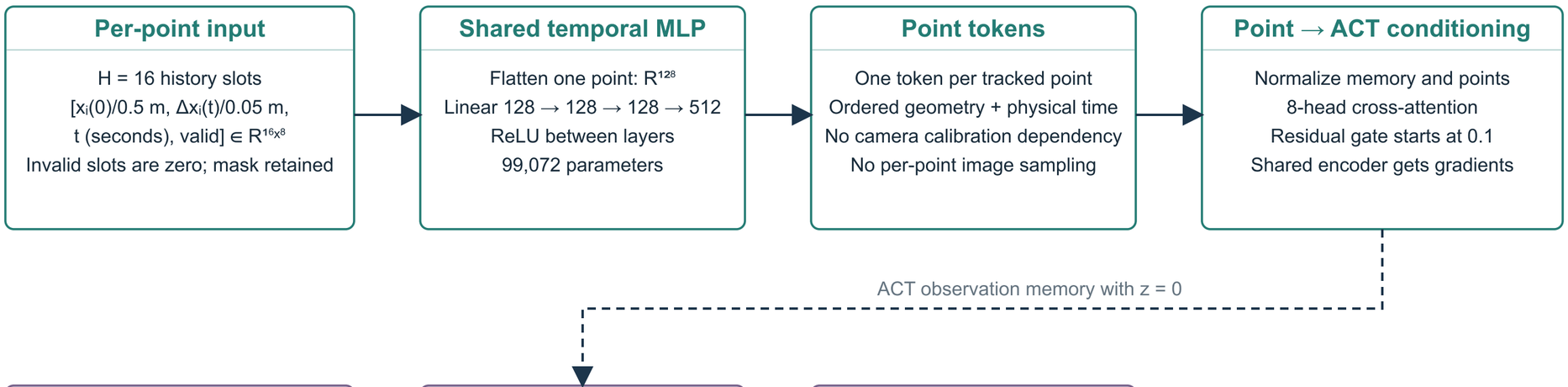
Inside the point branch

One trajectory MLP + one point-to-memory attention; native ACT action decoder and head stay unchanged



K points have no learned ID or order embedding. Changing point order only changes token order.

Dynamics and past completion use separate heads.
Each query attends to the same observed memory.
No query self-attention: query chunks preserve outputs.

Only the dynamics head also reads committed actions.
Completion gradient is detached or explicitly norm-capped.

Controlled comparison: stationary tracks vs observed tracks

Both branches use the same current anchors, timestamps, validity, visual backbone and parameter budget. Only numeric displacement is changed. No image gathering is present in v2.